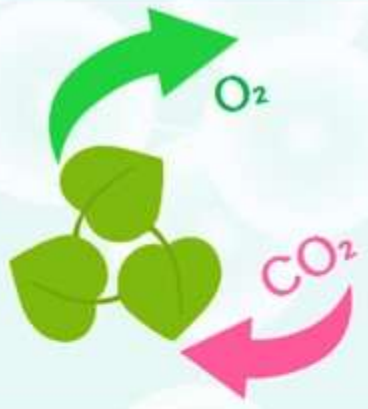


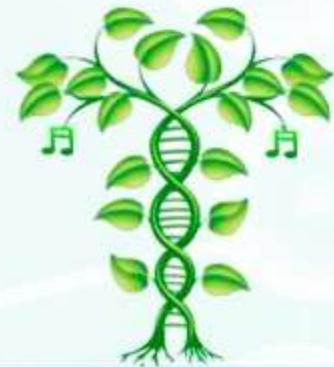


BIOLOGY



INFECTIOUS DISEASES

MODULE - 5



HOW WILL YOU KNOW IF YOU ARE ACTUALLY HEALTHY ?



PHYSICALLY FIT



SOCIALLY FIT



MENTALLY FIT

Health is defined as a state of physical, mental and social well being.



HOW WILL YOU KNOW IF YOU HAVE A DISEASE?



Disease is the condition when the functioning of one or more organ or system of the body is adversely affected and characterized by various symptoms



DISEASES- BASED ON TRANSMISSION

NON - COMMUNICABLE

- i. Gradual onset
- ii. Multiple causes
- iii. Long natural history
- iv. Prolonged treatment schedule
- v. Prolonged follow up
- vi. Care more than cure.

COMMUNICABLE

- i. Sudden onset
- ii. Single cause
- iii. Short natural history
- iv. Short treatment schedule
- v. Short follow up
- vi. Cure can be achieved.

Communicable disease is also called as infectious disease



DISEASES – CLASSIFICATION BASED ON FREQUENCY



1 ENDEMIC

The disease that exhibits a relatively steady frequency in a particular geographic locale over a long period of time

2 SPORADIC

Some occasional cases of diseases are reported at regular intervals

3 EPIDEMIC

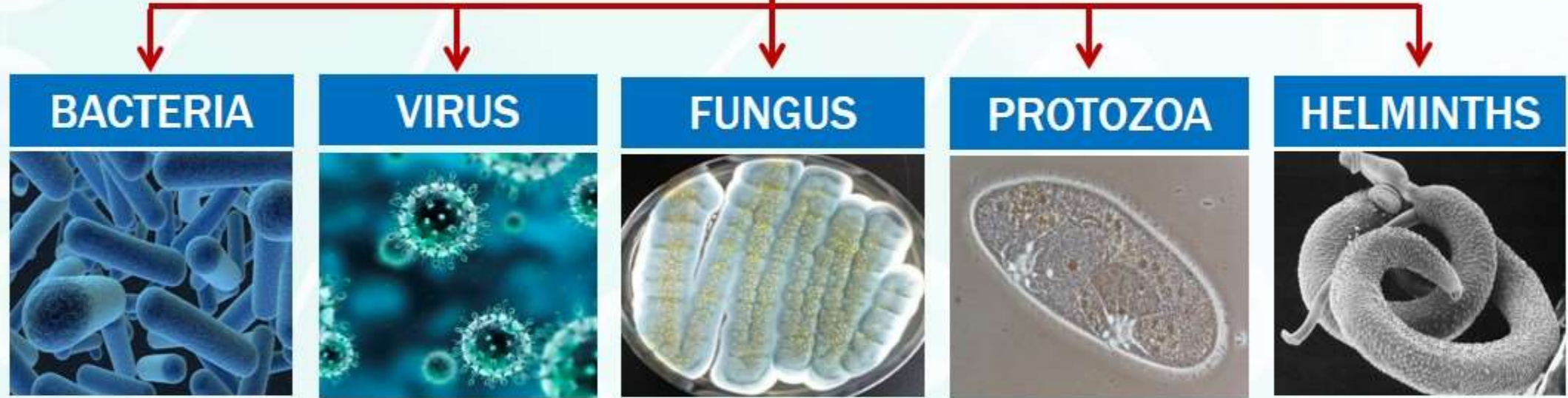
The disease becomes prevalent and the no. of cases increases rapidly beyond expected

4 PANDEMIC

When an epidemic disease is spread across continents.



INFECTIOUS MICROBES



Pathogen

Any infectious agent that produces disease.

Pathology

The branch of medical science that deals with lab. Experiments of samples of body tissue.



INFECTIOUS MICROBES – MODE OF TRANSMISSION



DROPLETS



FAECES



SEXUAL



ORAL



VERTICAL



IATROGENIC



DIRECT CONTACT



VECTOR BORNE



BACTERIAL DISEASES



TYPHOID



TETANUS



PNEUMONIA



TUBERCULOSIS



CHOLERA



PLAGUE



DIPHTHERIA



BACTERIAL DISEASES

DISEASE	CAUSATIVE ORGANISM	SPREAD
Typhoid	<i>Salmonella typhi</i>	Fecal- oral transmission
Tetanus	<i>Clostridium tetani</i>	From cut or deep wounds
Pneumonia	<i>Streptococcus pneumoniae</i>	Droplet contact
Cholera	<i>Vibrio cholerae</i>	Contaminated water and food
Plague	<i>Yersinia pestis</i>	Direct contact or airborne
Diphtheria	<i>Corynebacterium diphtheria</i>	Respiratory droplets transmission
Tuberculosis	<i>Mycobacterium tuberculosis</i>	Exposure to infected droplet



VIRAL DISEASES



POLIOMYELITIS



EBOLA



AIDS



SWINE FLU



DENGUE



ROTAVIRUS



INFLUENZA



MEASLES



VIRAL DISEASES

DISEASE	CAUSATIVE ORGANISM	SPREAD
Poliomyelitis	<i>Polio virus</i>	Fecal - oral transmission
Ebola	<i>Ebola virus (filovirus)</i>	Direct contact
AIDS	<i>Human Immuno Deficiency Virus</i>	Direct contact/sexual transmission/ vertical/ iatrogenic transmission
Swine flu	<i>H1N1 and H1N2</i>	Inhaling respiratory droplets
Dengue	<i>Dengue fever virus (flavivirus)</i>	Vector borne by female <i>Aedes aegypti</i> mosquito
Rotavirus	<i>Rotavirus</i>	Contact with infected objects
Measles	<i>Measles virus</i>	Droplet transmission



MEDICAL TEST TO DETECT SOME COMMON INFECTIOUS DISEASE

TUBERCULOSIS

PLAGUE

DENGUE

AIDS

TYPHOID

Mantoux Tuberculin skin test

Microscopic examination

Real time PCR, Antibody test

ELISA Test

Widal test



PROTOZOAN DISEASES

DISEASE	CAUSATIVE ORGANISM	SPREAD
MALARIA	<i>Plasmodium sp.</i>	Vector borne – female Anopheles mosquito bites
AMEBIASIS	<i>Entamoeba histolica</i>	Infected food and water, feces to oral transmission
LEISHMANIASIS	<i>Leishmania sp.</i>	Female Phlebotomine sandflies
GIARDIASIS	<i>Giardia sp.</i>	Fecal- oral transmission.
AFRICAN SLEEPING SICKNESS	<i>Trypanosoma sp.</i>	Vector borne - Tsetse fly bites



FUNGAL DISEASES



RINGWORM



ATHLETE'S FOOT

Trychophyton, Epidermophyton and Microsporum causes these types of diseases, which can spread through direct contact with the infected person or using any items of the infected person



HELMINTHIC DISEASES

ASCARIASIS

Ascariasis lumbricoides
roundworm



FILARIASIS

Wuchereria bancrofti
roundworm



TAENIASIS

Taenia solium
tapeworm



COMMON PLANT DISEASES CAUSED BY MICROBES

DISEASE	CAUSATIVE AGENT	
CROWN GALL	<i>Agrobacterium tumefaciens</i> (Bacteria)	
CITRUS CANCER	<i>Xanthomonas</i> sp. (Bacteria)	
POWDERY MILDEW	<i>Podosphaera xanthii</i> (Fungi)	



COMMON PLANT DISEASES CAUSED BY MICROBES

DISEASE	CAUSATIVE AGENT	
LATE BLIGHT	<i>Phytophthora infestans</i> (Fungi)	
COMMON RUST	<i>Phragmidium spp</i> (Fungi)	
SMUTS	<i>Teliomycetae</i> class of fungus	



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